Kazi Ababil Azam

Mail   Website   LinkedIn   Github   Google Scholar

Work Experience

Graduate Research Assistant (GRA), George Mason University — *Hardware & System Security* — Aug 2026 to Present, Fairfax, Virginia, USA

Joined the MASS Lab (Mason Augmented/Virtual Reality and System Security Lab) under the supervision of Dr. Yicheng Zhang. Working on hardware security and GPU security.

Software Engineer I, Synesis IT PLC — *Software Development, React, Go, TypeScript* — Jun 2024 to Aug 2026, Dhaka, Bangladesh

Contributed to the development of "Convay", a video conferencing and collaborative workspace application. Notable contributions include development of front-end meeting features, the desktop-native version (Convay Desktop), implementation of in-meeting annotation, and offline file sync for Convay Drive.

Project Intern, ERA InfoTech Limited — *Natural Language Processing, Data Analysis, Python* — May 2023 to Jun 2023, Dhaka, Bangladesh

Worked on a month-long internship project focused on analysis of real-life business data applying natural language processing techniques.

Education

Doctor of Philosophy (Ph.D.) in Electrical & Computer Engineering, George Mason University (GMU) — Aug 2026 to Present, Fairfax, Virginia, USA

Bachelor of Science (B.Sc.) in Computer Science & Engineering, Bangladesh University of Engineering & Technology (BUET) — *CGPA: 3.67/4.00* — Apr 2019 to Jul 2024, Dhaka, Bangladesh

Research Experience

Towards Intelligent Traffic Signaling in Dhaka City Based on Vehicle Detection and Congestion Optimization — Jun 2023 to Jun 2024, NSysS 2024, 2023, arXiv

Undergraduate thesis under the supervision of Dr. A. B. M. Alim Al Islam.
Posters presented at NSysS 2023, NSysS 2024. Manuscript archived.

Devised a system architecture for traffic signal scheduling that incorporates successive implementations of vehicle detection and multi-objective optimization in a Raspberry Pi, specifically focused on the non-lane-based, heterogeneous traffic of Dhaka.

Implemented the proposed system architecture in a cost-effective manner and tested in a simulated environment and at a five-road intersection.

Prepared the NHT-1071 (Non-lane-based and Heterogeneous Traffic) dataset, consisting of images of Dhaka City traffic which involves the procurement of the images, the labeling of the vehicles, and benchmarking against state-of-the-art models. Applied in the context of traffic signaling for heterogeneous traffic observed in developing countries.

Tracing Users' Privacy Concerns Across the Lifecycle of a Romantic AI Companion — May 2025 to Jun 2026, arXiv

Under the supervision of Dr. Dipto Das and Dr. Imtiaz Karim.
Manuscript submitted to a conference.

Collected Reddit posts expressing privacy concerns related to romantic AI chatbot platforms though purposive sampling of relevant subreddits and AI-assisted filtering of posts.

Analyzed collected posts discovering four recurrent themes to highlight that privacy in romantic AI is best understood as an evolving socio-technical governance problem.

Pedestrian Trajectory Categorization and Analysis in AV Simulation — Jun 2024 to Jan 2026

Under the supervision of Dr. Golam Md Muktadir, Dr. A. B. M. Alim Al Islam, and Dr. Fahim Hasan Khan.

Analysis of real-life pedestrian trajectories from datasets in order to formulate metrics for generative trajectory models in autonomous vehicle simulations.

Categorizing trajectories found in public pedestrian trajectory datasets based on human annotation and shape perception of their visual structures through online survey.

Academic Projects

Contradictory, My Dear Watson: Extended *ML, Natural Language Inference* Github

A project extended from a Kaggle competition to detect contradictions in multilingual text data, including Bangla text, through rigorous preprocessing and fine-tuning.

SyncInc *Django, React, MUI, PostgreSQL* Github

A web-based project management software designed for organizations. Stack includes PostgreSQL as the database, Django for the backend, ReactJS with Material UI for the frontend, Django REST framework for API development, and Django Channels for real-time notifications.

Laser Security System With Arduino *Arduino, Cyber-Physical Systems* Github | YouTube

A project to control a laser security system with microcontrollers, where the ATMega32 unit manages the laser and the Arduino unit controls the alarm and overall coordination.

Ray Tracing and Illumination Using OpenGL *C++, OpenGL* Github

Implementation of an image generation pipeline using ray tracing and illumination techniques to create realistic images of geometric shapes (sphere, pyramid, cube) and a 2D plane using OpenGL.

Compiler *Bison, Flex, C* Github

Built a compiler from scratch including steps for creating a symbol table, comprising of lexical analysis, semantic analysis, and intermediate code generation.

Implementation of OS Features *xv6* Github

Implementation of different parts of an operating system, such as system calls, scheduler, thread synchronization, and memory management, on the xv6 OS (RISC-V architecture), according to lab assignments.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, x86 Assembly, MySQL, Bison, Flex, Bash, Go, TypeScript
Frameworks: React, Bootstrap, Django, Express.js, Next.js, Electron.js, Wails, FastAPI, PRAW
Technologies: LaTeX, Docker, Oracle DBMS, PostgreSQL, Git, Wireshark, Postman, OpenGL, ns-2
Libraries: OpenCV, PyAV, PyTorch, Keras, Pandas, Scikit-learn, SciPy, NLTK, Matplotlib, Seaborn, p5.js

Competitions

Champion Student Poster, 11th NSysS 2024 *Poster Presentation* Poster | Certificate
5th Place, IEEE CS, BUET presents GameJam 2023 *p5.js* Github

Leadership

Director of Events, BUET Cyber Security Club Link
Jun 2023 to Apr 2024

Promotion and organization of cybersecurity related events, seminars and both intra and inter-university Capture The Flag (CTF) competitions multiple times throughout the tenure.

Test Scores

TOEFL: 113/120 (Reading: 29/30, Listening: 30/30, Speaking: 27/30, Writing: 27/30)
GRE: 326.0/346.0 (Verbal: 157/170, Quantitative: 165/170, AWA: 4.0/6.0)

References

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